



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTROTECHNOLOGY
NQF LEVEL 2

21 November 2025

This marking guideline consists of 6 pages.

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QUESTION 1

- 1.1
1.1.1 True
1.1.2 True
1.1.3 False
1.1.4 False
1.1.5 True

(5 × 1) (5)

- 1.2 Given: $L = 1 \text{ km} = 1\,000 \text{ m}$, $d = 6 \text{ mm}$, $\rho = 1,72 \times 10^{-8} \Omega \cdot \text{m}$

$$A = \frac{\pi d^2}{4}$$

$$A = \frac{\pi(6 \times 10^{-6})^2}{4}$$

$$A = 28,274 \times 10^{-6} \text{ m}^2 \checkmark$$

$$R = \frac{\rho L}{A}$$

$$R = \frac{(1,72 \times 10^{-8}) \times 1\,000}{28,274 \times 10^{-6}} \checkmark$$

$$R = 0,608 \Omega \checkmark \quad (3)$$

- 1.3 1.3.1 The resistors are connected in parallel. (1)

$$1.3.2 \quad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \checkmark$$

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{5} + \frac{1}{6} \checkmark$$

$$\frac{1}{R_p} = \frac{7}{10} \checkmark$$

$$R_p = 1,429 \Omega \checkmark \quad (4)$$

$$1.3.3 \quad I_T = \frac{V_T}{R_T}$$

$$I_T = \frac{12}{1,429} \checkmark$$

$$I_T = 8,397 \text{ A} \checkmark \quad (2)$$

$$1.3.4 \quad P = V_T \times I_T$$

$$P = 12 \times 8,397 \checkmark$$

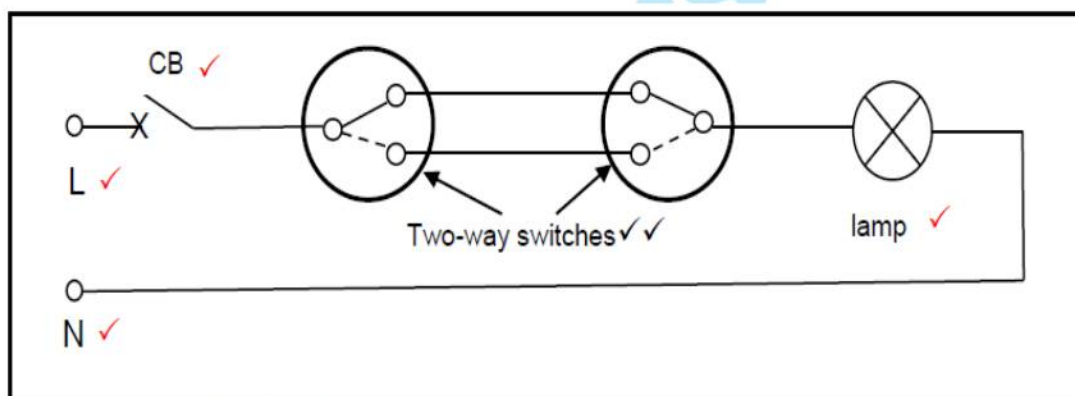
$$P = 100,764 \text{ W} \checkmark \quad (2)$$

- 1.4
- Lines of force always travel from the north pole to the south pole outside the magnet, and from the south pole to the north pole inside the magnet.
 - Each line of the magnetic flux forms a closed loop.
 - Magnetic lines never intersect.
 - Magnetic lines are like stretched elastic cords always trying to shorten themselves.
 - Lines of magnetic flux are parallel and in the same direction. (Any 3 × 1) (3)
- [20]

QUESTION 2

- 2.1
- | | |
|-------|----------------------|
| 2.1.1 | Double-pole isolator |
| 2.1.2 | Transformer |
| 2.1.3 | Push-button switch |
| 2.1.4 | Fuse |
| 2.1.5 | AC motor |
- (5 × 1) (5)

2.2



(6)

- 2.3
- Single-core PVC insulated cable
 - Flat twin and earth cable
 - Surfix cable
 - Bare copper earth wire
- (4)

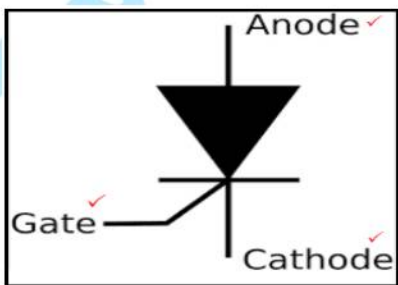
- 2.4
- | | |
|-------|--------------|
| 2.4.1 | 40 A |
| 2.4.2 | 20 A |
| 2.4.3 | 20 A or 30 A |
| 2.4.4 | 10 A |
- (4 × 1) (4)

- 2.5
- Earth bar
 - Neutral bar
- (2)

- 2.6
- Skirting is a material that lies along and protects and finishes the exposed edges of wireways. (1)

- | | | | |
|-----|-------|--|-------------|
| 2.7 | 2.7.1 | To cut conduit to the required length | |
| | 2.7.2 | To ensure that an enclosure is vertical or horizontally straight | |
| | 2.7.3 | To bend PVC conduit | |
| | | | (3 × 1) (3) |
| | | | [25] |

QUESTION 3

- | | | | |
|-----|-------|--|-----|
| 3.1 | 3.1.1 | Power supply unit (PSU) | (1) |
| | 3.1.2 | A – Transformer
B – Rectifier
C – Filter
D – Regulator/Voltage regulator | (4) |
| | 3.1.3 | The charge and discharge of the capacitor during the pulsating cycle produce a smoothed DC output. | (2) |
| 3.2 | | <ul style="list-style-type: none"> Only hold the soldering iron by the handle, or you will burn your hands. Store the soldering iron in a soldering iron stand. Be careful not to burn the electrical cord of the soldering iron. | (3) |
| 3.3 | | <ul style="list-style-type: none"> Switch off the circuit. Remove one leg of the resistor by desoldering one side. Connect the ohmmeter (or multimeter in ohm range) and read off the value. Compare this value to the colour code of the resistor, taking the tolerance into consideration. | (4) |
| 3.4 | | <ul style="list-style-type: none"> What needs to be done? Where does it have to be done? How will it be done? What resources are available? | (4) |
| 3.5 | |  | (3) |
| 3.6 | | <ul style="list-style-type: none"> Soldering iron Solder sucker Side cutters Long nose pliers | (4) |

[25]

QUESTION 4

- 4.1 4.1.1 B
 4.1.2 B
 4.1.3 D
 4.1.4 B
 4.1.5 B

(5 × 1) (5)

- 4.2 • Scope or purpose of the task
 • Safe work practices to be used
 • Personal protective equipment (PPE)
 • Use of insulating materials and tools
 • Special precautionary steps to take
 • Lubrication procedures
 • Repair procedures and inspections
 • Electrical diagrams
 • Equipment connections
 • Sketches and other pictorial materials
 • Data sheets

(Any 5 × 1) (5)

4.3 4.3.1 $R_P = \frac{R_2 \times R_3}{R_2 + R_3}$
 $R_P = \frac{5 \times 10}{5 + 10} \checkmark$
 $R_P = 3,333 \Omega \checkmark$

 $R_T = R_1 + R_P$
 $R_T = 2 + 3,333$
 $R_T = 5,333 \Omega \checkmark$

 $I_T = \frac{V_T}{R_T} = \frac{12}{5,333} = 2,25 A \checkmark$

(4)

4.3.2 $R_T = R_1 + R_3$
 $R_T = 2 + 10$
 $R_T = 12 \Omega \checkmark$
 $I_T = \frac{V_T}{R_T} = \frac{12}{12} = 1 A \checkmark$

(2)

4.3.3 $V_1 = I \times R$
 $V_1 = 1 \times 2 \checkmark$
 $V_1 = 2 V \checkmark$

(2)

- 4.3.4 • The total resistance will increase (R_1 plus R_3).
 • The current will decrease (current is inversely proportional to resistance).

(2)

[20]

QUESTION 5

ABC ELECTRIC 225 Botha Ave, Amp Park 2025		
		DATE
Customer department	CSG cafeteria✓	2 April 2025✓
Customer phone	072 345 6789✓	
Work request	Stove maintenance✓	
Work done	Check isolator, check conductor size, check earth bonding, check voltage at stove, check circuit breaker size✓	
Remarks	No earth continuity. Install the earth from the stove to the DB.✓	
Time of arrival	09:20✓	
Time of departure	11:20✓	
Work done by	Student name✓	SUPERVISOR: Mr Dlamini ✓

[10]

TOTAL: 100